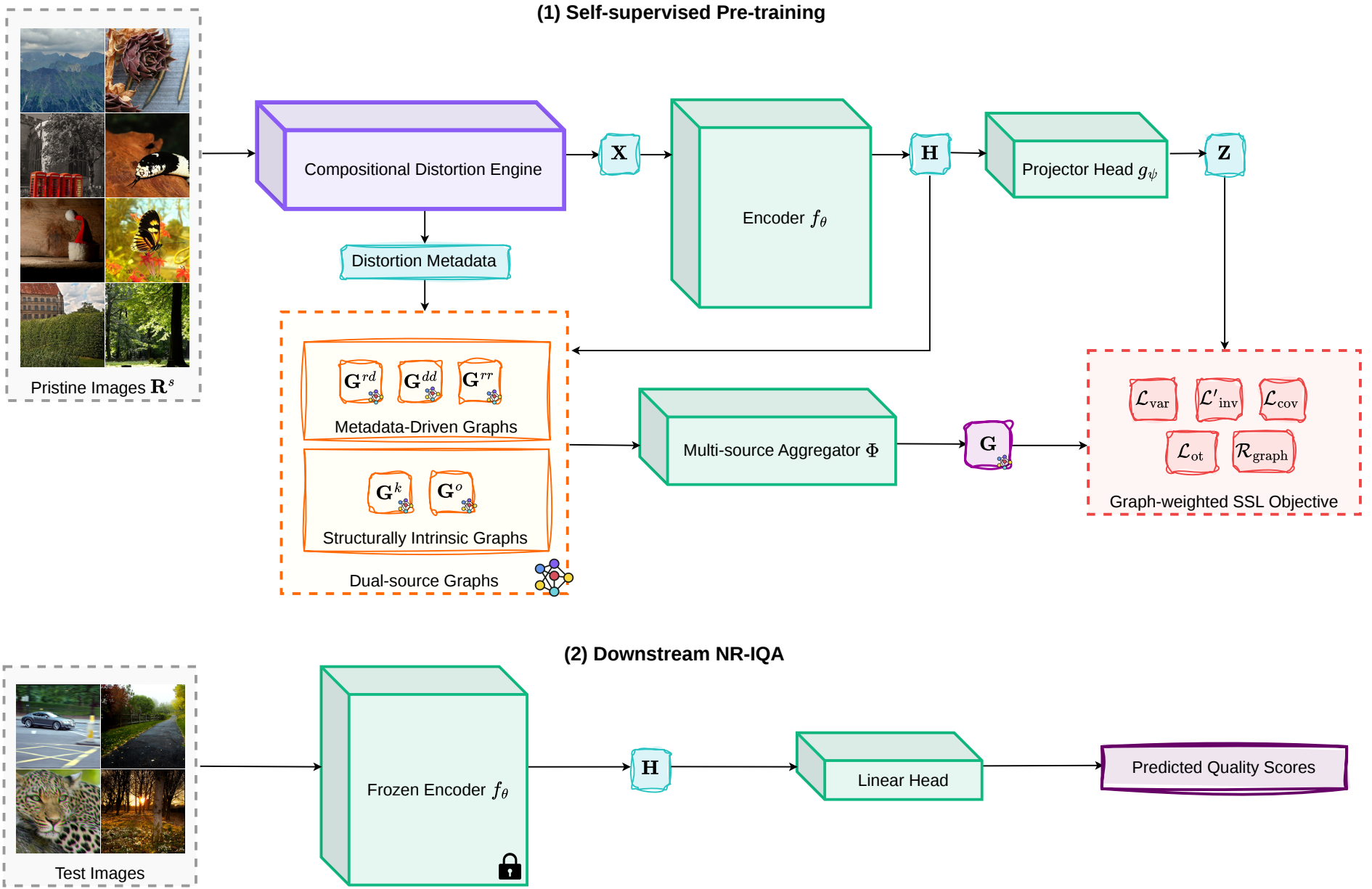
(1) Self-supervised Pre-training
Pristine Images $\mathbf{R}^s$
Compositional Distortion Engine
$\mathbf{X}$
Encoder f_θ
$\mathbf{H}$
Projector Head g_ψ
$\mathbf{Z}$
Distortion Metadata
$\mathbf{G}^{rd}$
$\mathbf{G}^{dd}$
$\mathbf{G}^{rr}$
Metadata-Driven Graphs
$\mathbf{G}^{k}$
$\mathbf{G}^{o}$
Structurally Intrinsic Graphs
Dual-source Graphs
Multi-source Aggregator Φ
$\mathbf{G}$
$\mathcal{L}_{\text{var}}$
$\mathcal{L}'_{\text{inv}}$
$\mathcal{L}_{\text{cov}}$
$\mathcal{L}_{\text{ot}}$
$\mathcal{R}_{\text{graph}}$
Graph-weighted SSL Objective
(2) Downstream NR-IQA
Test Images
Frozen Encoder f_θ
$\mathbf{H}$
Linear Head
Predicted Quality Scores